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SOME NONINTERIOR CONTINUATION METHODS FOR LINEAR COMPLEMENTARITY PROBLEMS*

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Abstract. We introduce some new path-following methods for the solution of the linear complementarity problem. We call these methods noninterior continuation methods since, in contrast to interior-point methods, not all iterates have to stay in the positive orthant. This is possible since we reformulate certain perturbed complementarity problems as a nonlinear system of equations. However, similar to interior-point methods, we also try to follow the central path. We present some conditions which guarantee the existence of this central path, prove a global convergence result for some implementable noninterior continuation methods, and report some numerical results obtained with these methods. We also prove global error bound results for the perturbed linear complementarity problems.

Key words. linear complementarity problems, path-following methods, interior-point methods, global error bounds, P_0-matrix, R_0-matrix

AMS subject classification. 90C33

1. Introduction. The linear complementarity problem, denoted by LCP(q, M), is to find a vector $z = (x, y) \in \Re^{2n}$ such that

$$x \geq 0, y \geq 0, x^T y = 0, y = Mx + q,$$

where $M \in \Re^{n \times n}$ is a given matrix and $q \in \Re^n$ is a given vector. This problem serves as a unified formulation of linear and quadratic programming problems as well as of two-person (noncooperative) matrix-games and has several important applications in economics and engineering sciences; see Cottle, Pang, and Stone [3], Harker [15], and Isac [17] for some examples.

There exist several methods for solving LCP(q, M); the interested reader is referred to the excellent books of Murty [31] and Cottle, Pang, and Stone [3]. Here we focus on the interior-point approach. This approach solves (approximately) a sequence of certain perturbed linear complementarity problems, PLCP(q, M, μ) for short; these perturbed problems depend on a positive parameter $\mu > 0$ and consist of finding a vector $z(\mu) = (x(\mu), y(\mu)) \in \Re^{2n}$ satisfying the conditions

$$x > 0, y > 0, x_i y_i = \mu \ \ (i \in I), y = Mx + q.$$

Here and throughout the paper, the index set I is an abbreviation for the set $\{1, \ldots, n\}$. Under certain conditions, the PLCP(q, M, μ) are uniquely solvable for all $\mu > 0$, and the pair $(\mu, z(\mu))$ forms a smooth trajectory, usually called the *central path.* By tracing this trajectory as μ tends to zero, one hopes to find a solution of the original LCP(q, M). Actually, this can be shown under suitable assumptions; see, e.g., Meggido [30], Kojima, Meggido, and Noma [23, 25, 26], and Kojima et al. [24].

The aim of this paper is to introduce some new tools to allow us to reformulate the PLCP(q, M, μ) as a nonlinear system of equations and to show that these tools can be used both for the theoretical analysis of a continuation method and for the

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numerical solution of LCP(q, M). These tools are defined in §2. In §3, we prove some results concerning the existence and uniqueness of the solution $z(\mu)$ of PLCP(q, M, μ). We also compare our approach with a recent method of Chen and Harker [2], which is very similar to our approach; actually, we show in §4 that their method can be viewed as a scaled variant of one of our methods. Some global error bound results are given in §5. In §6 we describe in detail the implemented algorithm and present a global convergence result for this method. The suggested algorithm differs from interior-point methods in at least two points: on the one hand, the iterates do not necessarily have to stay in the positive orthant, and on the other hand we can start with an arbitrary vector $(x^0, y^0) \in \Re^{2n}$ and an arbitrary initial parameter $\mu_0 > 0$. Some promising numerical results are given in §7. We conclude with some final remarks in §8.

Notation. For a vector $x = (x_1, \ldots, x_n)^T \in \Re^n$, the inequalities $x \geq 0$ and $x > 0$ are defined componentwise. If $x, y \in \Re^n$, the vector $z := (x^T, y^T)^T \in \Re^{2n}$ is usually abbreviated by $z = (x, y)$. Given two vectors $x, y \in \Re^n$, the symbol $\min\{x, y\}$ denotes the n-vector having $\min\{x_i, y_i\}$ as its ith component. All norms are Euclidean norms. By $dist(x, \mathcal{S})$ we denote the (Euclidean) distance of a vector $x \in \Re^n$ to a set $\mathcal{S} \subseteq \Re^n$; i.e., $dist(x, \mathcal{S}) = \inf_{y \in \mathcal{S}} \|x - y\|$.

2. Main tools. The noninterior continuation methods to be presented in the following section are based on reformulations of the PLCP as nonlinear systems of equations. The main tools used in these reformulations are functions $\varphi_\mu : \Re^2 \to \Re$ having the property; cf. [20].

$$\varphi_\mu(a, b) = 0 \Longleftrightarrow a > 0, b > 0, ab = \mu, \tag{1}$$

where, unless otherwise stated, μ is any fixed positive parameter. In this section, we introduce some functions φ_μ having this property; cf. [20].

LEMMA 2.1. *The function*

$$\varphi_\mu(a, b) := a + b - \sqrt{(a - b)^2 + 4\mu}$$

has the property (1).

Proof. First assume that $a > 0, b > 0$, and $ab = \mu$. Then, we obtain

$$\begin{aligned} \varphi_\mu(a, b) &= a + b - \sqrt{a^2 - 2ab + b^2 + 4ab} \\ &= a + b - \sqrt{(a + b)^2} \\ &= a + b - |a + b| \\ &= 0. \end{aligned}$$

To prove the converse result, assume that $\varphi_\mu(a, b) = 0$; i.e.,

$$a + b = \sqrt{(a - b)^2 + 4\mu} > 0. \tag{2}$$

Squaring both sides of the equation in (2), we get $ab = \mu$. Therefore $\text{sign}(a) = \text{sign}(b)$. Consequently it follows from the inequality in (2) that $a > 0$ and $b > 0$. □

LEMMA 2.2. *The function*

$$\varphi_\mu(a, b) := a + b - \sqrt{a^2 + b^2 + 2\mu}$$

has the property (1).

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Proof. If $a > 0, b > 0, ab = \mu$, we get

$$\begin{aligned}\varphi_\mu(a,b) &= a + b - \sqrt{(a+b)^2} \\ &= a + b - |a+b| \\ &= 0.\end{aligned}$$

On the other hand, the condition $\varphi_\mu(a,b) = 0$ can be rewritten as

$$a + b = \sqrt{a^2 + b^2 + 2\mu} > 0,$$

from which $a > 0, b > 0$, and $ab = \mu$ follow in a similar way as in the proof of Lemma 2.1. □

Remark. For the special case $\mu = 0$, the function introduced in Lemma 2.1 reduces to the min function used, e.g., by Pang [32, 33] and Harker and Pang [16], in order to characterize the complementarity problem itself as a (nonsmooth) system of equations. On the other hand, the function φ_μ defined in Lemma 2.2 coincides for $\mu = 0$ with a recently introduced function of Fischer [8] which has subsequently been used by several authors, including Fischer [9, 10, 11], Qi and Jiang [36], Facchinei and Soares [5, 6], Tseng [40], and Kanzow [19, 21].

Remark. Let φ_μ denote the function defined in Lemma 2.1 or 2.2. Then φ_μ is continuously differentiable for all $(a,b) \in \Re^2$, and it is not difficult to see that the partial derivatives have the property

$$\frac{\partial \varphi_\mu}{\partial a}(a,b), \frac{\partial \varphi_\mu}{\partial b}(a,b) \in (0,2)$$

for all $a, b \in \Re$. This property turns out to be important in the following section; see Theorem 3.5.

Remark. We have found two other functions having the property (1); namely,

$$\varphi_\mu(a,b) := \frac{1}{2}\min^2\{0, a+b\} - ab + \mu, \tag{3}$$

$$\varphi_\mu(a,b) := (a-b)^2 - a|a| - b|b| + 2\mu. \tag{4}$$

It is straightforward to see that these functions satisfy condition (1). However, they have at least two main disadvantages when comparing them with the functions of Lemmas 2.1 and 2.2. On the one hand, the functions (3) and (4) are more nonlinear than their counterparts of Lemmas 2.1 and 2.2; on the other hand, they do not have the property mentioned in the previous remark. We note that the functions (3) and (4) also reduce to some known functions for $\mu = 0$; see, e.g., Mangasarian [28], Kanzow [18, 19], and Kanzow and Kleinmichel [22].

There might exist several other functions φ_μ having the property (1). It is our feeling, however, that the two functions given in Lemmas 2.1 and 2.2 are the most interesting ones, both from a theoretical and a numerical point of view.

3. Continuation methods. Let $\mu > 0$. Throughout this section, let $\varphi_\mu : \Re^2 \to \Re$ denote the function given in Lemma 2.1 or 2.2. If we introduce the nonlinear operator $F_{\varphi_\mu} : \Re^{2n} \to \Re^{2n}$ defined by

$$F_{\varphi_\mu}(z) := F_{\varphi_\mu}(x,y) := \begin{pmatrix} Mx + q - y \\ \varphi_\mu(x,y) \end{pmatrix},$$

where

$$\varphi_\mu(x,y) := (\varphi_\mu(x_1,y_1), \ldots, \varphi_\mu(x_n,y_n))^T \in \Re^n,$$

we obtain as a direct consequence of Lemmas 2.1 and 2.2 the following characterization of PLCP(q, M, μ).

THEOREM 3.1. *A vector $z(\mu) := (x(\mu), y(\mu)) \in \Re^{2n}$ is a solution of PLCP(q, M, μ) if and only if $z(\mu)$ satisfies the nonlinear equation $F_{\varphi_\mu}(z) = 0$.*

We now investigate the following (theoretical) continuation method. A variant of this algorithm that can be implemented is described in §6.

ALGORITHM 3.2.

(S.0): Choose $z^0 := (x^0, y^0) \in \Re^{2n}$, set $k := 0$, and let $\{\mu_k\} \subseteq \Re$ be a strictly decreasing sequence with $\lim_{k\to\infty} \mu_k = 0$.

(S.1): Terminate if $z^k = (x^k, y^k)$ solves LCP(q, M).

(S.2): Find a solution $z^{k+1} := z(\mu_{k+1})$ of the nonlinear system of equations

$$F_{\varphi_{\mu_{k+1}}}(z) = 0$$

(or, equivalently, of PLCP(q, M, μ_{k+1})).

(S.3): Set $k := k + 1$ and go to (S.1).

In order for Algorithm 3.2 to be well defined, one has to guarantee that the perturbed problems have a (unique) solution z^{k+1}. A well-known condition which guarantees the existence and uniqueness of such a solution is that M is positive semidefinite and that there exists a strictly feasible vector $\hat{x}$ (i.e., $\hat{x} > 0$ and $M\hat{x} + q > 0$); see, e.g., Meggido [30] and Kojima, Mizuno, and Noma [26]. These assumptions have been weakened by Kojima et al. [24, Condition 2.1], where the following conditions are assumed to hold:

CONDITION KMNY.

(a) M is a P_0-matrix (see below).

(b) There exists a strictly feasible vector for LCP(q, M).

(c) The level sets $\mathcal{L}_t(q, M) := \{(x, y) \in \Re^n \mid x \geq 0, y \geq 0, y = Mx + q, x^T y \leq t\}$ are bounded for all $t \geq 0$.

Here we show that Algorithm 3.2 is well defined if M is a P_0- and R_0-matrix. (At the end of this section, we show how this assumption is related to Condition KMNY.) We first restate the definitions of these matrix classes as well as the definition of P-matrices.

DEFINITION 3.3. *A matrix $M \in \Re^{n\times n}$ is said to be a*

(a) *P_0-matrix $:\Longleftrightarrow \forall x \in \Re^n, x \neq 0, \exists i \in I : x_i \neq 0,$ and $x_i[Mx]_i \geq 0$;*

(b) *P-matrix $:\Longleftrightarrow \forall x \in \Re^n, x \neq 0, \exists i \in I : x_i \neq 0,$ and $x_i[Mx]_i > 0$;*

(c) *R_0-matrix $:\Longleftrightarrow$ LCP$(0, M)$ has $z^* := (0, 0) \in \Re^{2n}$ as its unique solution.*

It is not difficult to see that every P-matrix is both a P_0-matrix and an R_0-matrix. Moreover, it is well known that positive definite matrices are P-matrices and positive semidefinite matrices are P_0-matrices; see, e.g., Murty [31].

Our analysis of Algorithm 3.2 is similar to the one of Chen and Harker [2]. We stress, however, that we do not have to assume that all diagonal elements of M are positive. This contrasts favourably with Chen and Harker's method and is, in fact, the main advantage of our approach. We note that this positiveness assumption is often not satisfied; e.g., if we formulate a (linear or) quadratic programming problem as an LCP, the corresponding matrix M has zero entries on its diagonal (in case of a linear programming problem, all diagonal entries are zero).

Now we come to the convergence analysis of Algorithm 3.2. Consider the set of paths

$$\mathcal{P}(\bar{\mu}) := \{(\mu, z(\mu)) \mid 0 < \mu \leq \bar{\mu}, z(\mu) \text{ solves PLCP}(q, M, \mu)\},$$

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where $\bar{\mu} > 0$ is some given constant. The following theorem is due to Chen and Harker [2, Thm. 3.2] and follows from the implicit function theorem and simple continuity arguments.

THEOREM 3.4. *If the Jacobian matrix $\nabla F_{\varphi_\mu}(z(\mu))$ is nonsingular for all $z(\mu), 0 < \mu \leq \bar{\mu}$, then $\mathcal{P}(\bar{\mu})$ consists solely of continuously differentiable paths. If, in addition, $\mathcal{P}(\bar{\mu})$ is bounded, then every limit point of $z(\mu)$ as μ approaches zero is a solution of LCP(q, M).*

We next show that the Jacobian matrices $\nabla F_{\varphi_\mu}(z)$ are nonsingular for P_0-matrix LCPs.

THEOREM 3.5. *Assume that $M \in \Re^{n\times n}$ is a P_0-matrix. Then the Jacobian matrix $\nabla F_{\varphi_\mu}(z)$ is nonsingular for all $z \in \Re^{2n}$ and all $\mu > 0$.*

Proof. Let $z = (x, y) \in \Re^{2n}$ and $\mu > 0$ be fixed. Then, using the notation

$$D_a := D_a(z) := \operatorname{diag}\left(\ldots, \frac{\partial \varphi_\mu}{\partial a}(x_i, y_i), \ldots\right),$$

$$D_b := D_b(z) := \operatorname{diag}\left(\ldots, \frac{\partial \varphi_\mu}{\partial b}(x_i, y_i), \ldots\right),$$

the Jacobian matrix is given by

$$\nabla F_{\varphi_\mu}(z) = \nabla F_{\varphi_\mu}(x, y) = \begin{pmatrix} M & -I \\ D_a & D_b \end{pmatrix}. \tag{5}$$

Let $\nabla F_{\varphi_\mu}(z)p = 0$ with $p = \left(p^{(1)}, p^{(2)}\right), p^{(i)} \in \Re^n, i = 1, 2$. Then

$$Mp^{(1)} - p^{(2)} = 0, \tag{6}$$

$$D_a p^{(1)} + D_b p^{(2)} = 0. \tag{7}$$

From the second remark in §2, we have that the diagonal matrices D_a and D_b are positive definite. Thus, (7) can be rewritten as

$$p^{(2)} = -D_b^{-1} D_a p^{(1)}, \tag{8}$$

where $D_b^{-1} D_a$ is also a positive definite diagonal matrix. Inserting (8) into (6) yields

$$(M + D_b^{-1} D_a) p^{(1)} = 0. \tag{9}$$

Since M is a P_0-matrix and $D_b^{-1} D_a$ is positive definite, we directly obtain from Definition 3.3 that the matrix $M + D_b^{-1} D_a$ is a P-matrix. In particular, this matrix is nonsingular. Consequently, we have

$$p^{(1)} = 0.$$

Thus, we get

$$p^{(2)} = 0$$

from (8), which proves the theorem. □

The following theorem can be shown in a similar way as the corresponding theorem of Chen and Harker [2, Thm. 3.7].

THEOREM 3.6. *Let $M \in \Re^{n\times n}$ be a P_0-matrix, let $z^0 \in \Re^{2n}$, and assume that the level sets*

$$\mathcal{L}(\mu) := \{z \in \Re^{2n} \mid \|F_{\varphi_\mu}(z)\| \leq \|F_{\varphi_\mu}(z^0)\|\}$$

are uniformly bounded for all $0 < \mu \le \bar{\mu}$ *for some* $\bar{\mu} > 0$. *Then the PLCP*(q, M, μ) *have a unique solution* $z(\mu)$ *for all* $0 < \mu \le \bar{\mu}$.

The following is a (necessary and) sufficient condition for the above level sets to be bounded.

THEOREM 3.7. *Assume that* $M \in \Re^{n\times n}$ *is an* R_0*-matrix,* $z^0 \in \Re^{2n}$, *and* $\bar{\mu} > 0$. *Then the level sets* $\mathcal{L}(\mu)$ *as defined in Theorem* 3.6 *are uniformly bounded for all* $0 \le \mu \le \bar{\mu}$.

Proof. Assume there exists an unbounded sequence $\{z^k\} = \{(x^k, y^k)\} \subseteq \Re^{2n}$ such that $z^k \in \mathcal{L}(\mu_k)$ where $\{\mu_k\}$ is a sequence with $0 < \mu_k \le \bar{\mu}$ for all k. Since the mapping $\mu \to F_{\varphi_\mu}(z^0)$ is continuous and $[0, \bar{\mu}]$ is a compact interval, the maximum $\alpha := \max_{\mu\in[0,\bar{\mu}]} \|F_{\varphi_\mu}(z^0)\|$ exists. We therefore have

$$\|Mx^k + q - y^k\| \le \|F_{\varphi_{\mu_k}}(z^k)\| \le \|F_{\varphi_{\mu_k}}(z^0)\| \le \alpha.$$

Let $z^* := (x^*, y^*)$ be an accumulation point of the bounded sequence $\left\{\frac{(x^k,y^k)}{\|(x^k,y^k)\|}\right\}$. Then we obtain

$$0 \le \frac{\|Mx^k + q - y^k\|}{\|(x^k, y^k)\|} \le \frac{\alpha}{\|z^k\|} \to 0;$$

i.e., $z^* = (x^*, y^*)$ satisfies the equation

$$Mx^* = y^*. \tag{10}$$

Similarly, if φ_μ denotes one of the functions defined in Lemma 2.1 or 2.2, we get from the definition of these functions and the boundedness of the sequence $\{\mu_k\}$ that

$$\frac{\varphi_{\mu_k}(x_i^k, y_i^k)}{\|(x^k, y^k)\|} \to \varphi_0(x_i^*, y_i^*)$$

and, on the other hand,

$$\frac{|\varphi_{\mu_k}(x_i^k, y_i^k)|}{\|(x^k, y^k)\|} \le \frac{\alpha}{\|z^k\|} \to 0,$$

so that

$$\varphi_0(x_i^*, y_i^*) = 0$$

holds for all $i \in I$. This is equivalent to $x^* \ge 0, y^* \ge 0$, and $(x^*)^T y^* = 0$. In view of (10) this means that z^* is a solution of LCP$(0, M)$. However, since $\|z^*\| = 1$ and therefore $z^* \ne 0$, we have a contradiction to the assumed R_0-property of the matrix M. □

Using similar arguments as in Fischer [9] and Tseng [40], one can show that even the single level set $\mathcal{L}(\mu)$ for $\mu = 0$ is bounded for all $q \in \Re^n$ if and only if M is an R_0-matrix. Hence, the assumption of M being an R_0-matrix is also necessary for Theorem 3.7 to be true.

We can summarize the above results as follows: If $M \in \Re^{n\times n}$ is a P_0- and R_0-matrix, then all PLCP(q, M, μ) have a unique solution, the sequence $\{z^k\}$ generated by Algorithm 3.2 is bounded, and every limit point of this sequence is a solution of LCP(q, M). In particular, we obtain again by our constructive approach the result of Aganagic and Cottle [1] that the LCP has a nonempty solution set if M is a P_0- and R_0-matrix.

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We now turn back to the condition KMNY. Of course, the assumption of M being a P_0- and R_0-matrix is not directly related to condition KMNY since the former is independent of the specific vector q, whereas the latter depends on this vector. However, based on the previous results, we are now able to prove the following equivalence theorem.

THEOREM 3.8. *M is a P_0- and R_0-matrix if and only if condition KMNY is satisfied for all $q \in \Re^n$.*

Proof. First assume that M is a P_0- and R_0-matrix. Then condition KMNY (a) is obviously satisfied. Let $\mu > 0$. Since M is a P_0- and R_0-matrix, there exists a (unique) solution $z(\mu) = (x(\mu), y(\mu)) \in \Re^{2n}$ of the PLCP(q, M, μ). Obviously, $x(\mu)$ is then a strictly feasible vector for LCP(q, M); i.e., condition KMNY (b) is also satisfied. Finally, the validity of condition KMNY (c) follows from Proposition 3.9.23 in [3]. Conversely, assume that condition KMNY is satisfied for all $q \in \Re^n$. We only have to show that M is an R_0-matrix. Assume this is not true. Then there exists a nonzero solution $z = (x, y) \in \Re^{2n}$ of LCP$(0, M)$. Then the vector τz is also a solution of LCP$(0, M)$ for all $\tau \geq 0$. In particular, the level set $\mathcal{L}_t(0, M)$ is not bounded for any $t \geq 0$, which contradicts condition KMNY (c). □

We note that the main results of this section are now a simple consequence of Theorem 3.8 and of Theorem 4.4 in [24]. However, the proof of Theorem 3.8 (namely the existence of a strictly feasible vector) is based on the previous results of this section which will also play a central role in the following sections. On the other hand, Kojima et al. [24] were able to prove, under their condition KMNY, that the entire sequence $z(\mu)$ converges to a solution of LCP(q, M) as μ approaches zero. Hence we get the following corollary from Theorem 3.8 which improves on our Theorem 3.4.

COROLLARY 3.9. *If M is a P_0- and R_0-matrix, then the PLCP(q, M, μ) have a unique solution $z(\mu)$ for all $\mu > 0$, and the entire sequence $z(\mu)$ converges to a solution of LCP(q, M) as μ tends to zero.*

A similar result holds for Chen and Harker's method [2] which also improves on their Theorem 3.2.

4. The method of Chen and Harker. First note that it would also have been possible to use the tools of §2 to characterize the problem

$$x > 0, Mx + q > 0, x_i[Mx + q]_i = \mu \ \ (i \in I), \tag{11}$$

which is obviously equivalent to PLCP(q, M, μ). Using the functions φ_μ of Lemmas 2.1 and 2.2, respectively, we obtain the following characterizations of (11):

$$x_i + [Mx + q]_i - \sqrt{(x_i - [Mx + q]_i)^2 + 4\mu} = 0 \ \ (i \in I) \text{ and} \tag{12}$$

$$x_i + [Mx + q]_i - \sqrt{x_i^2 + [Mx + q]_i^2 + 2\mu} = 0 \ \ (i \in I). \tag{13}$$

On the other hand, Chen and Harker [2] give the following reformulation of (11):

$$m_{ii}x_i + [Mx + q]_i - \sqrt{(m_{ii}x_i - [Mx + q]_i)^2 + 4m_{ii}\mu} = 0 \ \ (i \in I). \tag{14}$$

This formulation is very similar to our characterization (12), but it has the disadvantage that all diagonal entries of M must be positive (otherwise (14) is not equivalent to (11)). If this is the case, however, it is also possible to reformulate problem (11) as follows:

$$m_{ii}x_i + [Mx + q]_i - \sqrt{(m_{ii}x_i)^2 + [Mx + q]_i^2 + 2m_{ii}\mu} = 0 \ \ (i \in I). \tag{15}$$

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This formulation can be regarded as the counterpart to (13). Note that the characterizations (12) and (14) coincide if all diagonal elements of M are equal to one. Under this assumption, the reformulations (13) and (15) are also equivalent.

Now let $S \in \Re^{n\times n}$ be any positive definite diagonal matrix. Then

$$Sy \geq 0 \Longleftrightarrow y \geq 0 \tag{16}$$

holds for any vector $y \in \Re^n$. Hence the LCP(q, M) is equivalent to

$$x \geq 0, S(Mx+q) \geq 0, x^T(Mx+q) = 0, \tag{17}$$

and interior-point-like methods for problem (17) try to solve the corresponding perturbed problem

$$x > 0, S(Mx+q) > 0, x_i[Mx+q]_i = \mu \ \ (i \in I). \tag{18}$$

If M has positive diagonal entries and if we define $S := \text{diag}(\ldots, 1/m_{ii}, \ldots)$, then the matrix $M_S := SM$ has unit entries on its diagonal. Consequently, Chen and Harker's characterization for this scaled problem is exactly our characterization (12) for this problem; i.e., Chen and Harker's method can be regarded as a scaled variant of one of our methods. Therefore, it is quite natural to apply our tools of §2 to the scaled problem (17)/(18). In §7, we consider three scaling matrices:

$$\begin{aligned} &S_0 := I_n \qquad \text{(no scaling)},\\ &S_1 := \text{diag}(s_1, \ldots, s_n), s_i := \begin{cases} 1/m_{ii} & \text{if } m_{ii} \neq 0,\\ 1 & \text{if } m_{ii} = 0,\end{cases}\\ &S_2 := \text{diag}(\sigma_1, \ldots, \sigma_n), \sigma_i := 1/\|M_{i.}\|, \end{aligned}$$

where $M_{i.}$ denotes the ith row of the matrix M. (Numerically the conditions $m_{ii} = 0$ and $m_{ii} \neq 0$ should, of course, be replaced by something like $|m_{ii}| < \tau$ and $|m_{ii}| \geq \tau$ for a small constant $\tau > 0$.)

We note, however, that the above scaling is somewhat strange and has its disadvantages; for example, we solve LCP(q_S, M_S) instead of LCP(q, M), where $q_S := Sq, M_S := SM$, and S is the scaling matrix, but some matrix-classes are not invariant under this (one-sided) scaling; see, e.g., Todd [37] for a brief discussion. Nevertheless, this scaling seems to be natural in view of the correspondence of our methods to Chen and Harker's characterization of PLCP(q, M, μ) and leads to a substantial improvement of the numerical results presented in §7.

5. Global error bounds for PLCP(q, M, μ). Consider the LCP in the following formulation: find a vector $x \in \Re^n$ satisfying the conditions

$$x \geq 0, Mx+q \geq 0, x^T(Mx+q) = 0.$$

The corresponding perturbed problem is exactly problem (11). Obviously, these problems are equivalent to LCP(q, M) and PLCP(q, M, μ), respectively, as defined in §1 (just take $y = Mx+q$), so we denote these problems also by LCP(q, M) and PLCP(q, M, μ), and the solution set of PLCP(q, M, μ) will again be denoted by $\mathcal{S}(\mu)$.

Although there are several existing error bound results for LCP(q, M)(see, e.g., [27, 29]), there is, to our knowledge, no error bound result for the PLCP(q, M, μ). Such a result, however, will play an important role in the algorithm to be described in the following section, namely in the termination rule of the inner iteration. We first give the relevant definitions.

DEFINITION 5.1. *Let $\mu > 0$, let $r : \Re^n \to \Re$, and assume that PLCP(q, M, μ) has a nonempty solution set $\mathcal{S}(\mu)$.*

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(a) *The continuous function r is called a* residual *of PLCP(q, M, μ) if $r(x) \geq 0$ for all $x \in \Re^n$ and $r(x) = 0$ if and only if x solves PLCP(q, M, μ).*

(b) *A residual r is a* lower local error bound *for PLCP(q, M, μ) if there exist constants $\tau_1 > 0$ and $c_1 > 0$ such that*

$$\tau_1 r(x) \leq dist(x, \mathcal{S}(\mu))$$

for all $x \in \Re^n$ with $r(x) \leq c_1$.

(c) *A residual r is a* lower global error bound *for PLCP(q, M, μ) if there exists a constant $\tau_1 > 0$ such that*

$$\tau_1 r(x) \leq dist(x, \mathcal{S}(\mu))$$

for all $x \in \Re^n$.

(d) *A residual r is an* upper local error bound *for PLCP(q, M, μ) if there exist constants $\tau_2 > 0$ and $c_2 > 0$ such that*

$$dist(x, \mathcal{S}(\mu)) \leq \tau_2 r(x)$$

for all $x \in \Re^n$ with $r(x) \leq c_2$.

(e) *A residual r is an* upper global error bound *for PLCP(q, M, μ) if there exists a constant $\tau_2 > 0$ such that*

$$dist(x, \mathcal{S}(\mu)) \leq \tau_2 r(x)$$

for all $x \in \Re^n$.

Let φ_μ denote one of the functions defined in Lemma 2.1 or 2.2. For $x \in \Re^n$, let $y = Mx + q$ and define $r_{\varphi_\mu}(x) := \|\varphi_\mu(x, y)\| = \|F_{\varphi_\mu}(x, y)\|$, where $\varphi_\mu(x, y) := (\varphi_\mu(x_1, y_1), \ldots, \varphi_\mu(x_n, y_n)) \in \Re^n$ and where F_{φ_μ} denotes the operator as introduced at the beginning of §3. In view of Lemmas 2.1 and 2.2, the two possible functions r_{φ_μ} are residuals for PLCP(q, M, μ). In this section we prove that these two functions provide both lower and upper global error bounds for PLCP(q, M, μ) if the matrix M is a P_0- and R_0-matrix.

We first show that r_{φ_μ} provide lower global error bounds for PLCP(q, M, μ). Note that this result holds for an arbitrary matrix $M \in \Re^{n \times n}$.

LEMMA 5.2. *Let $\mu > 0$ and assume that PLCP(q, M, μ) has a nonempty solution set $\mathcal{S}(\mu)$. Then there exists a constant $\tau_1 > 0$ such that $\tau_1 r_{\varphi_\mu}(x) \leq dist(x, \mathcal{S}(\mu))$ for all $x \in \Re^n$.*

Proof. We first note that r_{φ_μ} are globally Lipschitz-continuous functions; this follows immediately from the second remark in §2 and the integral mean value theorem. Let κ_1 denote the Lipschitz constant of r_{φ_μ}. Let $x \in \Re^n$ be arbitrary, and let $x(\mu)$ be the closest solution of PLCP(q, M, μ) to x. Then we have

$$r_{\varphi_\mu}(x) = |r_{\varphi_\mu}(x) - r_{\varphi_\mu}(x(\mu))| \leq \kappa_1 \|x - x(\mu)\| = \kappa_1 dist(x, \mathcal{S}(\mu)).$$

The assertion therefore follows by taking $\tau_1 = 1/\kappa_1$. □

Next we note that the two possible functions r_{φ_μ} provide upper local error bounds.

LEMMA 5.3. *Let $\mu > 0, M \in \Re^{n \times n}$ be a P_0-matrix, and $\mathcal{S}(\mu)$ be nonempty. Then there exist constants $\tau_2 > 0$ and $c_2 > 0$ such that $dist(x, \mathcal{S}(\mu)) \leq \tau_2 r_{\varphi_\mu}(x)$ for all $x \in \Re^n$ with $r_{\varphi_\mu}(x) \leq c_2$.*

Proof. Since M is a P_0-matrix, the Jacobian $\nabla F_{\varphi_\mu}(z)$ is nonsingular for all $z = (x, y) \in \Re^{2n}, y = Mx + q$. In particular, $\nabla F_{\varphi_\mu}(z(\mu))$ is nonsingular for $z(\mu) = (x(\mu), y(\mu))$, where $x(\mu) \in \mathcal{S}(\mu)$ and $y(\mu) = Mx(\mu) + q$. Using Lemma 4.1.16 from

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[4] and the continuity of r_{φ_μ}, we therefore obtain the existence of $\kappa_2 > 0$ and $c_2 > 0$ such that

$$\begin{aligned} r_{\varphi_\mu}(x) &= \|F_{\varphi_\mu}(z)\| = \|F_{\varphi_\mu}(z) - F_{\varphi_\mu}(z(\mu))\| \geq \kappa_2\|z - z(\mu)\| \geq \kappa_2\|x - x(\mu)\| \\ &= \kappa_2 dist(x, \mathcal{S}(\mu)) \end{aligned}$$

for all $x \in \Re^n$ such that $r_{\varphi_\mu}(x) \leq c_2$, where $x(\mu)$ denotes the closest solution of x in $\mathcal{S}(\mu)$. From this the assertion follows by taking $\tau_2 := 1/\kappa_2$. □

We are now in a position to prove the main result of this section.

THEOREM 5.4. *Let $\mu > 0$ and let $M \in \Re^{n\times n}$ be a P_0- and R_0-matrix. Then there exists a constant $\tau_2 > 0$ such that $dist(x, \mathcal{S}(\mu)) \leq \tau_2 r_{\varphi_\mu}(x)$ for all $x \in \Re^n$; i.e., r_{φ_μ} provide upper global error bounds for PLCP(q, M, μ).*

Proof. Based on Lemma 5.3, the proof is similar to the one of Theorem 2.1 by Mangasarian and Ren [29]. For simplicity, we assume that φ_μ is the function from Lemma 2.2. (The proof is analogous for the function φ_μ of Lemma 2.1.) By Theorems 3.6 and 3.7, PLCP(q, M, μ) has a unique solution $x(\mu)$. Assume the theorem is false. Then there exists a vector $x^k \in \Re^n$ such that

$$\|x^k - x(\mu)\| > k r_{\varphi_\mu}(x^k) \tag{19}$$

for all k. Since r_{φ_μ} is an upper local error bound by Lemma 5.3, one can prove as in [29] that there exist $k_0 > 0$ and $\varepsilon > 0$ such that $r_{\varphi_\mu}(x^k) > \varepsilon$ for all $k \geq k_0$. From this and (19) we get $\|x^k\| \to \infty$. Let x^* be an accumulation point of the bounded sequence $\{\frac{x^k}{\|x^k\|}\}$. Note that $\|x^*\| = 1$ and therefore $x^* \neq 0$. Dividing both sides of (19) by $\|x^k\|$, we obtain

$$1 = \lim_{k\to\infty} \frac{\|x^k - x(\mu)\|}{\|x^k\|} \geq \lim_{k\to\infty} k\frac{r_{\varphi_\mu}(x^k)}{\|x^k\|}. \tag{20}$$

From the definition of φ_μ, the unboundedness of the sequence $\{x^k\}$ and the fact that $\frac{x^k}{\|x^k\|} \to x^*$ on a subsequence, we get

$$\varphi_\mu(x_i^k, y_i^k)/\|x^k\| \to x_i^* + [Mx^*]_i - \sqrt{(x_i^*)^2 + [Mx^*]_i^2} \quad (i \in I) \tag{21}$$

on this subsequence, where $y^k := Mx^k + q$. In view of (20), we see that the right-hand side of (21) is equal to 0 for all $i \in I$. This, however, means that $\varphi_0(x_i^*, [Mx^*]_i) = 0$ for all $i \in I$; i.e., x^* solves LCP$(0, M)$ (cf. the first remark in §2). Since $x^* \neq 0$, this contradicts the assumption that M is an R_0-matrix. □

We note that the constants τ_1 and τ_2 in Lemmas 5.2 and 5.3 and Theorem 5.4 depend on the matrix M and the value of the perturbation parameter $\mu > 0$ but not on the particular choice of the vector q.

6. Implemented algorithm and its convergence. The following algorithm is an implementable version of Algorithm 3.2. Instead of solving the nonlinear systems $F_{\varphi_\mu}(z) = 0$ of step (S.2) of Algorithm 3.2 exactly, we try to solve them inexactly using just one step of Newton's method for a fixed value of μ. If this step is successful (in a certain sense as defined below, cf. step (S.5) of Algorithm 6.1), we reduce the perturbation parameter μ; otherwise, we perform another Newton step for the same value of μ. Note that Newton's method is globalized by a line search for the merit function $\frac{1}{2}\|F_{\varphi_\mu}\|^2$.

ALGORITHM 6.1.

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(S.0): (Initial Data)
Let $\varphi_\mu : \Re^2 \to \Re$ be any of the functions defined in §2. Choose $x^0 \in \Re^n$, set $y^0 := Mx^0 + q, z^0 := (x^0, y^0)$, let $\mu_0 > 0, \beta \in (0,1), \sigma \in (0, \frac{1}{2}), \epsilon \geq 0, \gamma > 0, \eta \in (0,1)$, and set $k := 0$.

(S.1): (Termination Criterion)
If $err(z^k) := \| \min\{x^k, y^k\}\| \leq \epsilon$, stop: z^k is an (approximate) solution of LCP(q, M).

(S.2): (Computation of a Search Direction)
Compute $\Delta z^k \in \Re^{2n}$ as the solution of the linear system

$$\nabla F_{\varphi_{\mu_k}}(z^k)\Delta z = -F_{\varphi_{\mu_k}}(z^k).$$

(S.3): (Computation of a Steplength)
Let $t^k = \beta^{m_k}$, where m_k is the smallest nonnegative integer m satisfying the Armijo condition

$$\|F_{\varphi_{\mu_k}}(z^k + \beta^m \Delta z^k)\|^2 \leq (1 - \sigma\beta^m)\|F_{\varphi_{\mu_k}}(z^k)\|^2.$$

(S.4): (New Iterate)
Set $z^{k+1} := z^k + t^k \Delta z^k$.

(S.5): (Updating Rule for μ_k)
Define the vector $\varphi_{\mu_k}(x^{k+1}, y^{k+1}) := (\varphi_{\mu_k}(x_1^{k+1}, y_1^{k+1}), \ldots, \varphi_{\mu_k}(x_n^{k+1}, y_n^{k+1}))^T \in \Re^n$. If $\|\varphi_{\mu_k}(x^{k+1}, y^{k+1})\| \leq \gamma\mu_k$, then $\mu_{k+1} := \eta\mu_k$, else $\mu_{k+1} := \mu_k$.

(S.6): (Loop)
Set $k := k + 1$, and go to (S.1).

Remark. From our specific choice of the starting vector $z^0 = (x^0, y^0)$ and a simple induction argument, it follows that the relation $y^k = Mx^k + q$ holds for all iterations k. Note that such a relation does not hold in commonly used (infeasible) interior-point methods. Moreover, this motivates our termination criterion in step (S.1) and, in view of the global error bound results of the previous section, also the updating rule for μ_k in step (S.5).

Remark. There is now a strong theoretical foundation of the termination criterion used in step (S.1) of the above algorithm. From error bound results of Mangasarian and Ren [29] and Luo and Tseng [27] it follows that $err(z)$ is a global error bound of LCP(q, M) for all $q \in \Re^n$ if and only if $M \in \Re^{n\times n}$ is an R_0-matrix.

Remark. The linear system in step (S.2) of our algorithm has been solved as follows: Let $\varphi_{\mu_k}(x^k, y^k) := (\ldots, \varphi_{\mu_k}(x_i^k, z_i^k), \ldots)^T \in \Re^n$, and let D_a^k and D_b^k denote the diagonal matrices diag$(\ldots, \frac{\partial\varphi_{\mu_k}}{\partial a}(x_i^k, y_i^k), \ldots)$ and diag$(\ldots, \frac{\partial\varphi_{\mu_k}}{\partial b}(x_i^k, y_i^k), \ldots)$, respectively. Then, compute Δx^k from the $n \times n$ system

$$(D_a^k + D_b^k M)\Delta x^k = -\varphi_{\mu_k}(x^k, y^k) - D_b^k(Mx^k + q - y^k)$$

and set

$$\Delta y^k = M(x^k + \Delta x^k) + q - y^k.$$

We now show, using the theory developed in §3 for the theoretical Algorithm 3.2, that Algorithm 6.1 is also globally convergent under the same assumptions.

THEOREM 6.2. *Let $\varphi_\mu : \Re^2 \to \Re$ denote one of the functions defined in Lemma 2.1 or 2.2. Assume that $M \in \Re^{n\times n}$ is a P_0- and R_0-matrix, and let $\epsilon = 0$ (ϵ being the constant from the termination criterion). Let $\{z^k\}$ be any infinite sequence generated by Algorithm 6.1. Then*

(a) *the sequence $\{z^k\}$ is well defined;*

(b) *the sequence* $\{z^k\}$ *has at least one accumulation point*;

(c) *any accumulation point of* $\{z^k\}$ *is a solution of LCP(q, M).*

Proof. (a) Since M is a P_0-matrix, the linear systems in step (S.2) are uniquely solvable for all k by Theorem 3.5. Moreover, it is well known that the search direction $\Delta z^k \in \Re^{2n}$ obtained in this step is always a descent direction for the merit function $\frac{1}{2}\|F_{\varphi_{\mu_k}}\|^2$. Hence a steplength $t^k > 0$ can always be computed in step (S.3). Therefore Algorithm 6.1 is well defined.

(b) Since $\mu_k \in [0, \mu_0]$ for all k and since our algorithm is a descent method for any fixed μ_k, the entire sequence $\{z^k\}$ remains bounded in view of Theorem 3.7 and the assumed R_0-matrix property of M. Thus the sequence $\{z^k\}$ has at least one accumulation point.

(c) We first show that the sequence $\{\mu_k\}$ generated by Algorithm 6.1 converges to zero. Assume the contrary; i.e., assume there exists an iteration index $\hat{k}$ such that $\mu_{\hat{k}} = \mu_{\hat{k}+l}$ for all $l = 1, 2, 3, \ldots$. This means that Algorithm 6.1 eventually reduces to a damped Newton method for the (single) system of nonlinear equations $F_{\varphi_{\mu_{\hat{k}}}}(z) = 0$. In view of Theorems 3.5–3.7, however, it is well known that this method converges to the unique solution $z(\mu_{\hat{k}})$ of this system. In particular, the condition in step (S.5), i.e.,

$$\|\varphi_{\mu_k}(x^{k+1}, y^{k+1})\| = \|\varphi_{\mu_{\hat{k}}}(x^{k+1}, y^{k+1})\| \le \gamma\mu_k = \gamma\mu_{\hat{k}},$$

is satisfied for a finite value $k \ge \hat{k}$, so that the parameter μ_k will be reduced in this step. Hence $\{\mu_k\}$ converges to zero.

Now let $\bar{z} = (\bar{x}, \bar{y})$ be one of the accumulation points of $\{z^k\}$. (Because of (b), such an accumulation point exists.) Let $\{z^{k+1}\}_{k\in K}$ denote a subsequence converging to $\bar{z}$. For each sufficiently large $k \in K$, let $l(k)$ denote the largest index for which the condition $\|\varphi_{\mu_k}(x^{k+1}, y^{k+1})\| \le \gamma\mu_{l(k)}$ is satisfied; i.e.,

$$l(k) := \max\{j \in \{1, \ldots, k\} \mid \|\varphi_{\mu_k}(x^{k+1}, y^{k+1})\| \le \gamma\mu_j\}.$$

We note that this index $l(k)$ is well defined for all k sufficiently large and that $\lim_{k\in K} \mu_{l(k)} = 0$ since μ_k is reduced infinitely many times according to the first part of this proof. (Note that the index $l(k)$ is not necessarily in the index set K.) Taking into account the definition of $l(k)$ as well as the continuity of the two possible functions φ_μ in their arguments and in the perturbation parameter μ, we have

$$\|\varphi_0(\bar{x}, \bar{y})\| = \lim_{k\in K} \|\varphi_{\mu_k}(x^{k+1}, y^{k+1})\| \le \gamma \lim_{k\in K} \mu_{l(k)} = 0;$$

i.e., $\varphi_0(\bar{x}, \bar{y}) = 0$. This, however, is equivalent to $\bar{x}_i \ge 0, \bar{y}_i \ge 0$, and $\bar{x}_i\bar{y}_i = 0$ $(i \in I)$. According to the first remark after Algorithm 6.1, we also have $\bar{y} = M\bar{x} + q$. Hence $\bar{z} = (\bar{x}, \bar{y})$ is a solution of LCP(q, M). □

The implemented version of Algorithm 6.1 differs in two points from the above description, namely in steps (S.3) and (S.5). Instead of step (S.3), we employ the following nonmonotone Armijo rule (see Grippo, Lampariello, and Lucidi [13]):

(S.3') Define $p_k := \min\{k, p\}$ and let $t^k := \beta^{m_k}$, where m_k is the smallest nonnegative integer m satisfying the nonmonotone Armijo condition

$$\|F_{\varphi_{\mu_k}}(z^k + \beta^m \Delta z^k)\|^2 \le \max_{j=k-p_k+1,\ldots,k} \|F_{\varphi_{\mu_j}}(z^j)\|^2 - \sigma\beta^m \|F_{\varphi_{\mu_k}}(z^k)\|^2.$$

Here, p is any fixed nonnegative integer. In our numerical experiments, this nonmonotone line search gives better results than the standard (monotone) Armijo rule. This,

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we think, is an interesting aspect since usually the nonmonotone line search is only preferred for highly nonlinear objective functions, whereas our merit function is not too nonlinear.

The second modification is in the updating rule for the perturbation parameter μ_k. The updating rule used is the following one. It is very similar to the one presented by Chen and Harker [2] and works very well in our numerical experiments.

UPDATING RULE FOR μ_k:

(a) Let $u_{k+1} := err(z^{k+1})^2/n$. If $u_{k+1} \geq 1$, then $\mu_{k+1} := \sqrt{u_{k+1}}$, else $\mu_{k+1} := u_{k+1}$.

(b) If $\mu_{k+1} > \mu_k$, set $\mu_{k+1} := \mu_k$.

(c) If $\|\varphi_{\mu_k}(x^{k+1}, y^{k+1})\| < 10^{-4}$, set $\mu_{k+1} := 10^{-1}\mu_{k+1}$.

(d) If $\mu_{k+1} < 10^{-16}$, then $\mu_{k+1} := 10^{-16}$.

The main part of this (heuristic) updating rule is part (a). Part (b) guarantees that the sequence $\{\mu_k\}$ is nonincreasing; part (c) means that the parameter μ_k is reduced very fast if we are already close to a solution of LCP(q, M). Part (d), of course, is just a safeguard.

7. Numerical results. In this section, we present numerical results for the following methods:

1: This is the algorithm described in §6 with φ_μ being the function defined in Lemma 2.1.

2: This is the corresponding algorithm with φ_μ taken from Lemma 2.2.

3: Similar to methods 1 and 2 but with φ_μ from (3).

4: The same as method 3, but with φ_μ as defined in (4).

Furthermore, we will consider the three scaling strategies as suggested at the end of §4. We compare our results with the following two algorithms:

CH 1: This is Chen and Harker's method; cf. (14).

CH 2: This is the modification of Chen and Harker's method as suggested in (15). (The body for the algorithms CH 1 and CH 2 is, of course, the same as for the methods 1–4; the main difference is only in step (S.2) of Algorithm 6.1, where different linear systems are to be solved.) The algorithms have been implemented in MATLAB and tested on a 486 PC-type computer. The parameters used are as follows:

$$\beta = 0.5, \sigma = 10^{-4}, \epsilon = 10^{-6}, p = 10, \tau = 10^{-8}.$$

The initial barrier parameter μ_0 is $\|q\|/n$.

The first two test examples are well scaled, and we therefore present numerical results only for the choice $S = S_0$ of the scaling matrix.

EXAMPLE 7.1 (see Murty [31]). n variable,

$$M = \begin{pmatrix} 1 & 2 & 2 & \cdots & 2 \\ 0 & 1 & 2 & \cdots & 2 \\ 0 & 0 & 1 & \cdots & 2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}, q = (-1, \ldots, -1)^T.$$

This example of an LCP is a standard test problem for which both Lemke's complementary pivot algorithm and Cottle and Danzig's principal pivoting method are known to run in exponential time; see Murty [31, Chap. 6]. Its solution is $x^* = (0, \ldots, 0, 1)^T, y^* = (1, \ldots, 1, 0)^T$. Obviously, the matrix M in this example is a P-matrix and therefore a P_0- and R_0-matrix. Consequently, methods 1 and 2 satisfy

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TABLE 1
Number of iterations for Example 7.1.

Method	$n = 8$	$n = 16$	$n = 32$	$n = 64$	$n = 128$	$n = 256$
1	7	7	7	5	5	5
2	7	7	7	6	5	5
3	10	10	10	11	12	12
4	10	10	10	11	12	12

TABLE 2
Number of iterations for Example 7.2.

Method	$n = 8$	$n = 16$	$n = 32$	$n = 64$	$n = 128$	$n = 256$
1	8	8	8	8	8	7
2	8	8	9	9	9	10
3	11	14	15	17	18	20
4	11	13	14	16	18	20
CH 1	8	11	11	11	11	11
CH 2	9	10	10	10	10	10

the assumptions of the global convergence Theorem 6.2. The starting vector chosen is $x^0 = (1, \ldots, 1)^T$. The number of iterations needed by our algorithm is indicated in Table 1 for several values of the dimension n. The methods CH 1 and CH 2 need the same number of iterations as the methods 1 and 2, respectively, since the diagonal entries of M are all equal to one, so that the corresponding characterizations of the complementarity problem coincide.

EXAMPLE 7.2 (see Fathi [7]). n variable,

$$M = \begin{pmatrix} 1 & 2 & 2 & \cdots & 2 \\ 2 & 5 & 6 & \cdots & 6 \\ 2 & 6 & 9 & \cdots & 10 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 2 & 6 & 9 & \cdots & 4(n-1)+1 \end{pmatrix}, q = (-1, \ldots, -1)^T.$$

This LCP(q, M) is a standard test problem too. Again, the complementary pivot algorithm and the principal pivoting method are known to run in exponential time. Moreover, some Newton-type algorithms are also known to have some difficulties with this example; see, e.g., Harker and Pang [16] and Kanzow [19]. Its solution is $x^* = (1, 0, \ldots, 0)^T, y^* = (0, 1, \ldots, 1)^T$. The matrix M of this example is positive definite and therefore a P-matrix. Our numerical results obtained with the methods 1–4, CH 1, and CH 2 are summarized in Table 2. The starting vector chosen is the same as in Example 7.1.

From Tables 1 and 2, we see that methods 1 and 2 as well as methods CH 1 and CH 2 lead to better results than methods 3 and 4. This is not an unexpected behaviour since methods 3 and 4 are based on the more nonlinear functions φ_μ from (3) and (4), hence Newton's method leads to more difficulties in solving the corresponding nonlinear systems of equations $F_{\varphi_\mu}(z) = 0$. In the following randomly generated examples, we will therefore concentrate on these four methods.

EXAMPLE 7.3 (see Harker and Pang [16]). The matrix M is computed as follows: let $A, B \in \Re^{n \times n}$ and $q, \eta \in \Re^n$ be randomly generated such that $a_{ij}, b_{ij} \in (-5, 5), q_i \in (-500, 500)$, and $\eta_i \in (0.0, 0.3)$ and that B is skew-symmetric. Define $M = A^T A + B + diag(\eta)$. Then, M is a P-matrix. For several values of n, 10 examples have been generated in this way. The maximum, average, and minimum numbers of iterations

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TABLE 3
Number of iterations for Example 7.3.

Method		1			2			CH 1	CH 2
n		S_0	S_1	S_2	S_0	S_1	S_2		
$n = 50$	Max.	19.0	12.0	12.0	20.0	12.0	12.0	16.0	16.0
	Avg.	15.0	10.4	10.3	16.8	10.8	10.7	14.7	14.8
	Min.	12.0	9.0	9.0	13.0	10.0	9.0	13.0	13.0
$n = 100$	Max.	20.0	13.0	13.0	20.0	12.0	13.0	16.0	18.0
	Avg.	17.3	11.2	11.4	18.2	11.2	11.4	15.2	16.3
	Min.	16.0	9.0	10.0	17.0	9.0	10.0	14.0	15.0
$n = 150$	Max.	22.0	13.0	14.0	22.0	13.0	14.0	18.0	21.0
	Avg.	18.6	11.4	11.6	19.6	11.4	11.6	16.4	18.2
	Min.	17.0	11.0	11.0	18.0	11.0	11.0	16.0	16.0
$n = 200$	Max.	22.0	14.0	15.0	21.0	14.0	15.0	18.0	19.0
	Avg.	18.9	11.7	11.8	19.6	12.0	12.1	16.5	18.1
	Min.	17.0	11.0	11.0	19.0	11.0	11.0	14.0	16.0

TABLE 4
Number of iterations for Example 7.4.

Method		1			2			CH 1	CH 2
n		S_0	S_1	S_2	S_0	S_1	S_2		
$n = 50$	Max.	19.0	13.0	13.0	21.0	13.0	13.0	17.0	18.0
	Avg.	16.0	11.1	11.1	17.6	11.3	11.6	14.3	14.9
	Min.	13.0	9.0	10.0	15.0	10.0	10.0	13.0	13.0
$n = 100$	Max.	20.0	12.0	12.0	24.0	12.0	12.0	17.0	18.0
	Avg.	19.0	10.8	11.0	20.9	10.3	10.5	15.6	16.7
	Min.	16.0	9.0	10.0	18.0	9.0	10.0	13.0	15.0
$n = 150$	Max.	23.0	12.0	13.0	26.0	13.0	14.0	18.0	19.0
	Avg.	20.6	11.3	11.2	23.1	11.3	11.5	16.9	18.3
	Min.	19.0	10.0	10.0	21.0	10.0	10.0	15.0	17.0
$n = 200$	Max.	24.0	13.0	12.0	25.0	13.0	13.0	19.0	20.0
	Avg.	21.1	11.4	11.2	23.0	11.6	11.9	17.2	18.4
	Min.	18.0	10.0	10.0	21.0	10.0	10.0	17.0	18.0

needed by the algorithms are summarized in Table 3. In all test runs, $x^0 = (0, \ldots, 0)^T$ has been chosen as starting vector.

EXAMPLE 7.4 (see Harker and Pang [16] "hard examples"). In this example, M is computed in the same way as in the previous example, and $q \in \Re^n$ is randomly generated with entries $q_i \in (-500, 0)$. Table 4 contains our numerical results, which we have obtained using the starting vector $x^0 = (0, \ldots, 0)^T$.

EXAMPLE 7.5 (see Pardalos et al. [34]). This example is a P_0-matrix LCP. The matrix M, the vector q, as well as the starting point x^0 are randomly generated. A detailed description is given in Pardalos et al. [34]. We note that at least half of the diagonal entries of M are zero, so that neither Chen and Harker's method nor the suggested modification are applicable to this example. Moreover, there are substantial difficulties in solving this example for the potential reduction algorithm of Pardalos et al. [34]. (Their algorithm needs hundreds of iterations in order to solve this problem.) On the other hand, as is shown in Table 5, our algorithms do not have any problems with this example.

We can summarize the above results as follows: All problems have been solved using only a small number of iterations. The number of iterations needed is almost independent of the problem dimension. Methods 1 and 2 lead to better results than do methods 3 and 4, and it is difficult to decide which of these two methods is best, although for most examples method 1 has a slightly better behaviour. In their un-

TABLE 5
Number of iterations for Example 7.5.

Method		1			2		
n		S_0	S_1	S_2	S_0	S_1	S_2
$n = 100$	Max.	17.0	17.0	17.0	17.0	16.0	20.0
	Avg.	14.8	10.7	8.3	16.5	13.7	14.2
	Min.	14.0	9.0	6.0	15.0	12.0	12.0
$n = 200$	Max.	19.0	12.0	13.0	19.0	15.0	21.0
	Avg.	16.4	10.7	9.2	17.5	13.8	16.1
	Min.	14.0	9.0	6.0	16.0	12.0	13.0

scaled versions, these two algorithms, however, are inferior to both Chen and Harker's method CH 1 and its modification CH 2. On the other hand, simple diagonal scalings of the original data lead to substantial improvement of the results obtained with the methods 1 and 2; these methods are by far the most successful ones. Both scaling techniques, S_1 and S_2, lead to similar results, so the slightly cheaper scaling S_1 might be preferable. We emphasize that we have not introduced the diagonal scaling in order to improve our results. (In fact, there might be better scaling techniques, though this is not a simple problem.) Instead the considerations of §4 show that it is more natural to compare our algorithms with the one of Chen and Harker after rescaling the original problem.

Finally, we note that in almost all iterations the full stepsize $t^k = 1$ has been accepted. (In just 10 of the 820 test runs made for our numerical results we observed a steplength less than 1; this happened 6 times in Example 7.2, 1 time in Example 7.3, and 3 times in Example 7.4.) This is not the case if the nonmonotone line search rule is replaced by a monotone one. Moreover, it is very unlikely that a practical interior-point method for LCP(q, M) will almost always take a full step, since the condition that all iterates have to stay in the positive orthant will usually truncate the steplength. Taking the stepsize $t^k = 1$ very often (in a controlled way), however, is now known to be very successful in combination with Newton-type methods; see, e.g., the highly promising and extensive numerical results reported by Grippo, Lampariello, and Lucidi [13, 14] and Toint [38, 39]. Another advantage of our noninterior continuation methods is the fact that we can start at an arbitrary vector $(x^0, y^0) \in \Re^{2n}$, in particular, the very natural choice $y^0 = Mx^0 + q$ is allowed even if y^0 has nonpositive or negative components. This contrasts favourably with interior-point methods.

8. Final remarks. In this paper, we have introduced some new continuation methods for the solution of LCPs. As with interior-point methods, we try to follow the central path; however, we also allow negative iterates. One of our methods is closely related to a recently proposed method of Chen and Harker [2], but here we do not need their assumption that all diagonal entries of M are positive.

There are several questions which remain to be answered: a theoretical justification of the updating rule for the perturbation parameter μ is missing. We also have not given a local or global rate of convergence result. Moreover, it is currently not known to the author whether or not the noninterior continuation methods have a polynomial complexity bound. The numerical results of the previous section seem to indicate that such a complexity bound exists since the number of iterations is almost independent of the problem dimensions.

Of course, the main tools introduced in §2 are also applicable to (convex) constrained optimization problems and could lead to noninterior continuation methods for, e.g., linear and quadratic programming problems. It would be interesting to com-

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pare these methods with standard interior-point methods as well as with the shifted barrier method of Freund [12] and the modified barrier method of Polyak [35]. The methods of Freund and Polyak also allow negative components of the iteration vectors and are, in this respect, related to our algorithms.

For the LCP, we plan to compare the numerical behaviour of the noninterior continuation methods suggested in this paper with some interior-point methods.

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